

STAT427

Problem Statement and Project Plan

Illinois Campus Cluster Usage Analysis

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Outline

- 1 Background
- 2 Project Overview and Client
- 3 Data
- 4 Problem and Plan
- 5 Deliverables

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What Is the Illinois Campus Cluster?

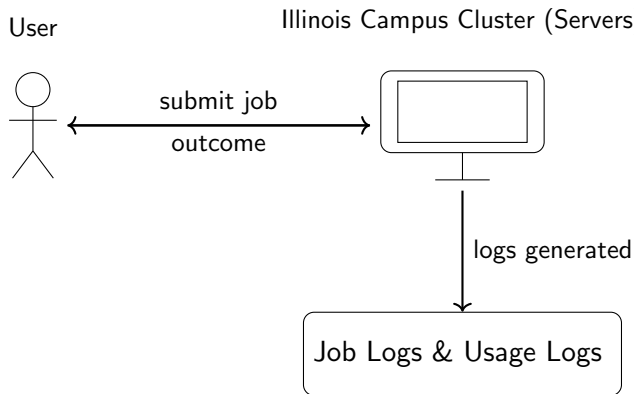
The **Illinois Campus Cluster** is a campus-wide high-performance computing resource.

- It provides shared computing and storage capacity to researchers across disciplines.
- Users submit their code and data as jobs, which are executed on the cluster's servers.
- During execution, the system automatically generates job and usage logs.



What Happens When a User Runs a Job on the Cluster?

Our analysis focuses on the **job logs** and **usage logs** generated by the cluster.



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Brief Project Overview

- The Illinois Campus Cluster supports over 3,000 active users per year
- The cluster serves a diverse research community with mixed funding sources
- Tons of data available on jobs and cluster usage, but no meaningful analysis has been performed on them.
- This consulting project focuses on summarizing and interpreting historical usage data to support operational decision-making

Client Introduction and Project Contacts

- **Primary Client Contact:** Christopher Heller
 - Director, National Center for Supercomputing Applications
 - Primary point of contact for project scope and overall goals
- **Secondary Stakeholder:** Jed Willard
 - Illinois Campus Cluster Program
 - Available for consultation, but not the primary point of contact
- **Data Contact:** Wyatt Charles Madej
 - Systems Engineer for the cluster
 - Primary contact for data access and data-related questions

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Two datasets are included for Jan '25 - Dec '25:

- **Job-Level Data**

- Generated from the SLURM workload manager and contains detailed job-level information
- Includes a large number of fields describing resource requests, execution, and completion
- Field definitions are documented in the SLURM job accounting documentation
- Preliminary dataset (~ 2.7 GB) is available for January to October and analysis-ready. The full dataset (> 5 GB) is also available, but stored in separate text files and uncleaned
- Additionally, another group of organized text data is provided with a focus on CPU utilization

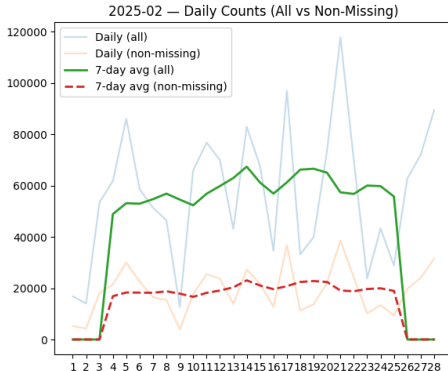
- **Cluster Usage Time Series**

- Aggregated cluster-level metrics stored in InfluxDB at 5-minute temporal resolution
- Includes information such as job queue size, running jobs, CPU/GPU utilization, allocated memory, and system-level measurements
- Access to these data is currently being coordinated with the client

Preliminary EDA

• Job-Level Data

- Every job has a clear submission date, even for jobs with missing user data
- Job distribution is generally roughly stationary
- Missing user/partition data is consistent



Preliminary EDA Continued

• Job-Level Text Data

- Most jobs use a lower number of cores
- More than half of the jobs use cores very inefficiently
- Several jobs utilize cores very efficiently, so the efficiency for this study is capped at 1

	cores	cpu_eff_capped	mem_used_mb
count	95460.0	92517.0	95460.0
mean	16.069118	0.440246	12903.78546
std	28.502183	0.361676	318190.492539
min	1.0	0.0	0.0
25%	1.0	0.062097	3.02
50%	7.0	0.415356	580.96
75%	16.0	0.75	4884.48
max	128.0	1.0	91970600.96

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Problem Statement and Ownership

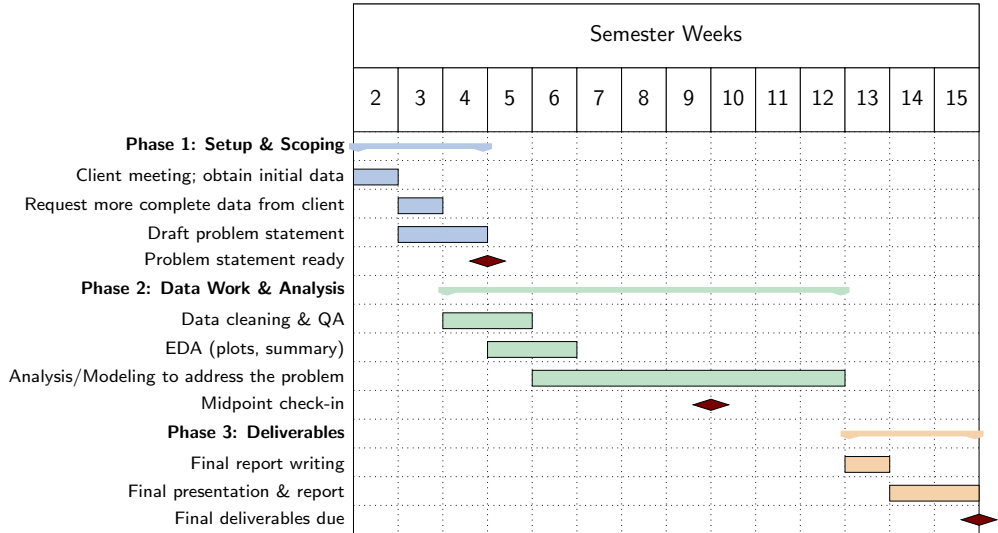
We will focus on the following topics:

- ① **User demographics and departmental usage patterns** $\Rightarrow$ Rushil
- ② **Types of workloads conducted on the cluster** $\Rightarrow$ Rishab
- ③ **Timing patterns of interactive versus batch jobs** $\Rightarrow$ Vlad
- ④ **Job submission behavior and execution schedules** $\Rightarrow$ Jiachen, Yanlin

Specific Analysis Questions

- 1 Analyze computational resource usage across user groups (e.g., efficiency and concentration of usage).
- 2 How do different types of workloads—categorized based on keywords in JobName—differ in terms of requested CPU and memory resources?
- 3 Analyze whether the job submission times differ across partitions, including patterns such as daytime vs nighttime or early vs late semester.
- 4 Analyze submission frequency across user groups; identify users with high submission frequency and long-running jobs, and evaluate whether these users tend to request more computational resources.
- 5 Bonus: Assess job queuing behavior by examining waiting times between submission and execution.

Project Timeline



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While determining final outputs is an ongoing discussion with our clients, we currently have the following goals:

- Analysis of usage across user groups
- Clustering model for automatically labeling job types
- Analysis of temporal patterns in cluster usage
- Framework for identifying under-/over-utilization in real time

Thank you for listening!

Questions?